Transformer-Based Object Detection in Low-Altitude Maritime UAV Remote Sensing Images

Zhiqiang Jiao, Min Wang, Sibo Qiao, Yanhan Zhang, and Zhanhua Huang

***Abstract*—Object detection technology plays an essential role in uncrewed aerial vehicle (UAV) sea search and rescue missions, which aim to quickly locate crucial targets such as trapped people and ships at sea in the complex marine environment. However, due to the restricted view angle of the UAV and the specificity of the working environment, the remote sensing images captured in the marine environment are characterized by small targets and considerable interference on the sea surface, which brings significant challenges to the UAV sea rescue mission. To address this issue, we propose an object detection model based on the Transformer architecture in this article. The model takes feature correction and double sampling detection transformer (FCDS-DETR) as the baseline and introduces a 2-D Gaussian probability density function as an additional attention mechanism to speed up the model's location of suspicious targets in the image to improve the detection accuracy of the model for targets. At the same time, the denoising training method is introduced in the model's training process to stabilize the bipartite graph matching and promote the convergence of the model. On the SDS ODv2 and AFO object detection datasets, our model achieves an average precision (AP) of 51% and 54.5%, respectively, which is an improvement of 5.7% and 4.4% compared to the baseline model's performance on these datasets.**

***Index Terms*—DETR, object detection, remote sensing images, transformer, uncrewed aerial vehicle (UAV).**

I. Introduction

With the continuous improvement in the performance of maritime uncrewed aerial vehicles (UAVs), it has become a reality to utilize UAV gimbal cameras to capture high-resolution, large field-of-view, and multiangle remote sensing images of the sea in a low-altitude state. Especially in maritime search and rescue missions, it is of great significance to use maritime low-altitude UAVs to capture remote sensing images of the target sea area and combine them with object detection algorithms to locate rescue targets accurately. However, due to the special characteristics of the working scenario and the maritime environment, the object detection task based on low-altitude maritime UAV remote sensing images faces some unique challenges, such as the small size of the target to be detected, the complexity and variability of the maritime background, and the variable lighting conditions [1].

To address these challenges, early researchers proposed various detection methods based on manual prior knowledge [2], [3], [4], achieving impressive performance in certain specific scenarios. For example, in the late 20th century, Sumimoto et al. [2] proposed a search algorithm focusing on color information, effectively detecting rescue targets by extracting regions of interest (ROI) of orange rescue targets, performing region segmentation, and database matching. Ramírez et al. [3] developed a maritime rescue system that collaborated with a UAV and an uncrewed surface vessel (USV). In this system, the UAV is responsible for location prediction and image search, while the USV updates real-time predictions, achieving efficient task allocation. Dinnbier et al. [4] introduced a method combining a dual-peak Gaussian mixture model (GMM) and fast Fourier transform (FFT). By removing dynamically changing backgrounds in images, this method significantly improved the detection performance and robustness of rescue targets.

These early studies demonstrated the importance and effectiveness of manually defined features based on prior knowledge in UAV-based maritime object detection tasks. However, solely relying on prior knowledge of image characteristics such as color, frequency, texture, and shape of the target has its limitations [5]. For example, in the face of complex and variable backgrounds and changing lighting conditions at sea, detection can be easily disrupted by factors such as waves, sea foam, reflective light, and shadows, leading to a decline in detection accuracy. Moreover, the robustness is also insufficient when dealing with small-sized targets and dynamic environmental changes.

In recent years, with the development of deep neural network technology, it has achieved subversive achievements in many fields. Especially in the field of aircraft control and image processing, such as adaptive control, object detection, image segmentation, and image generation [6], [7], deep neural networks play an increasingly important role. Researchers have also tried to apply this technique to the task of object detection in remote sensing images from maritime UAVs to improve object detection accuracy and model robustness in complex maritime environments. For example, Do Trong et al. [8] proposed a deep learning model that combines Faster-RCNN with the InceoptionNetv2 [9] network. The model used the adapted InceoptionNetv2 network as a backbone for feature

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extraction and was trained on a simulated marine environment dataset constructed by a virtual engine. Experimental results show that the model achieves effective results in recognizing victims in the ocean. However, the method has only been experimented on virtual datasets and has not been validated on real remote sensing datasets with complex backgrounds, which limits its reliability and effectiveness in practical applications. To overcome this limitation, Jayalath and Munasinghe [10] proposed the use of a pre-trained Fast-RCNN model for real-time UAV-based search and rescue tasks. The model was first pre-trained on the COCO [11] dataset and then fine-tuned on Okutama-Action [12], a remote sensing image dataset of a real maritime scene, to finally achieve precise localization of rescue targets at sea. However, although the method has a high detection accuracy, its choice to run the model on a low-performance Raspberry Pi edge device results in a time of 8 s per image capture and processing, which has a serious negative impact on the detection accuracy and speed of the model. To address this problem, Bai et al. [13] proposed a YOLOv5s-based object detection method for maritime rescue. The method uses YOLOv5s as a baseline model and improves the model's detection accuracy for targets by adding an extension layer. In addition, the Ghost module and C3Host module are introduced to realize a lightweight detection network and improve the detection speed of the model. However, target detection at sea is still not satisfactory in terms of accuracy. In addition, the above deep neural network-based target detection methods for maritime remote sensing images rely on the support of manual components, such as anchor and non-maximum suppression (NMS) [14], in the pre-processing or post-processing stage, which cannot realize the real end-to-end detection.

Aiming to overcome the defects of the above algorithms, this article proposes a novel end-to-end high-precision object detection method based on the Transformer architecture, which aims to improve the model's detection accuracy of targets at sea. Specifically, we introduce a Gaussian probability density distribution module into the baseline model [15] to dynamically modulate the search range of query vectors in cross-attention through Gaussian weight graphs, thus accelerating the model's location of suspicious targets in remote sensing images. In addition, we also introduce a denoising training method to overcome the slow convergence of DETR-like models. The specific contributions of this article are as follows.

1) First, we design a learnable Gaussian probability density distribution module in 2-D space, which utilizes the Gaussian probability density distribution to adjust the search range of the Transformer's query vectors in the cross-attention mechanism, thus improving the model's detection accuracy of maritime targets.
2) Second, to stabilize the bipartite graph-matching process of the baseline model, we introduce a denoising training method. This method prompts the model to reconstruct the real bounding box by injecting random noise into the bounding box and category labels, which speeds up the convergence speed and improves the detection accuracy.

This article is organized as follows. Section II reviews related research work. Section III provides a detailed description of the proposed method, including the network architecture, fusion strategy, and implementation details. Section IV presents the experimental setup, covering the dataset, evaluation metrics, comparative experiments, ablation studies, and result analysis. The conclusion and future directions are discussed in Section V.

II. Related Work

A. Generic Object Detection

Object detection is an important research direction in computer vision, which not only needs to recognize the types of various targets in an image but also determine the target's position in the image and label it using a bounding box. Traditional convolution-based object detection networks can be categorized into two-stage object detection and single-stage object detection. In the two-stage object detection algorithm, the first stage mainly utilizes a region proposal network (RPN) to extract the candidate regions where the target may appear. In the second stage, the classifier and regressor are utilized to classify and regress the targets in the candidate regions to complete the object detection process. Classical two-stage object detection networks include Fast-RCNN [16] and Faster-RCNN [17]. Unlike the RPN-based two-stage algorithm, the single-stage object detector completes the identification and localization of the target in a single stage. This improves the detection speed of the model, but usually at the expense of detection accuracy. The YOLO algorithm [18], [19], [20] and the SSD algorithm [21] are representative of the approaches.

In recent years, the Transformer architecture has achieved remarkable success in the field of natural language processing (NLP) [22], [23], [24]. As research continues, more and more researchers have begun to explore its application to computer vision tasks. Such attempts have shown some potential and promise. In particular, in the task of object detection, the Transformer can capture long-range dependencies, which enables the model to better understand the contextual information of the scene. DETR [25] is the first successful application of the Transformer in the field of object detection. Compared to traditional CNN-based object detection networks, DETR removes the RPN and anchor steps, making the structure of the model simpler and more uniform. However, DETR suffers from slow convergence and poor detection of small targets.

To solve this problem, we proposed a new object detection method called feature correction and double sampling detection transformer (FCDS-DETR) in our previous study. FCDS-DETR significantly improves the DETR model in terms of detection accuracy and convergence speed by introducing channel attention and a double sampling mechanism. Meanwhile, the model shows great potential in the detection of small and medium-sized targets.

Therefore, in this study, we use FCDS-DETR as our baseline model instead of other methods.

B. Object Detection in UAV Remote Sensing Images

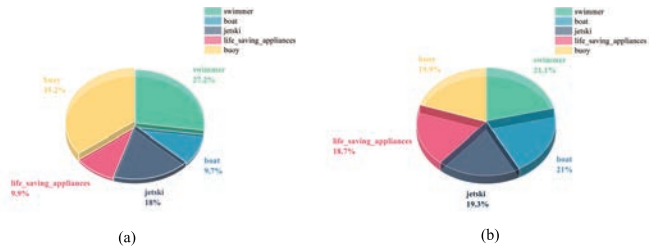
Directly applying the well-performing object detection model in the generic scene to the UAV remote sensing

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images, the results are often unsatisfactory [26]. To address this problem, researchers have proposed various solutions to enhance the performance of deep learning models for UAV object detection. For example, the use of multiscale feature fusion [27], [28], [29], improved backbone network [30], [31], and model pre-training [32], [33], etc. Among them, multiscale is generally implemented in CNN-based deep learning models by means of feature pyramid networks (FPNs) [34], [35]. FPNs capture both high-level semantic information and bottom-level detail information by efficiently building a hierarchical feature representation in a single network structure. In the object detection model of the Transformer architecture, the multiscale approach draws on the encoder structure of Deformable-DETR [36]. It obtains encoder output features of different sizes by inputting features of different scales extracted from the backbone network. Literature [37] improves the object detection performance of UAV images by introducing the last four scales of feature maps of the backbone network in YOLOv3 to obtain more texture and contour information. Li et al. [38] proposed the GeleNet network, which uses Transformer as its backbone structure to extract four-level feature embeddings with global dependencies. Combined with the D-SWSAM spatial attention module and the KTM knowledge transfer module, it achieves efficient fusion of local interactions and contextual information of features, thereby significantly improving the detection performance of salient objects in remote sensing images. Mao et al. [39] proposed HSF-DETR, a lightweight object detector based on Transformer. This method introduces a hybrid progressive fusion network, HPFNet, and a multiscale feature map module, effectively enhancing real-time object detection capabilities in UAV scenarios. Zhu et al. [40] proposed multispectral DETR, a multispectral remote sensing object detector based on a deformable attention mechanism, and introduced DropSpectrum and SwitchSpectrum strategies to fully exploit the complementary characteristics of visible light (RGB) and infrared (IR) images, thereby effectively improving the overall performance of remote sensing object detection. Ma et al. [41] proposed QETR, a query-enhanced Transformer object detector, which effectively alleviates the current Transformer architecture's insufficient local feature modeling capabilities in remote sensing image object detection tasks by introducing a query enhancement mechanism. At the same time, the proposed scaling control module utilizes the target shape variance information between adjacent layers of the decoder to enhance the model's ability to express targets in complex scenes while retaining multiscale target structural features, thereby significantly improving detection performance. Zhang et al. [42] further improved the model's detection accuracy for targets by replacing the Transformer decoder in their object detection model. In order to improve the detection performance of the model, the proposed object detection scheme in the maritime rescue scenario uses the above multiscale method based on Transformer architecture by default.

C. Data Enhancement

The size of the dataset has a large impact on the performance of the model and its generalization ability. In the field of low-altitude UAV remote sensing image object detection, there is still a lack of large datasets that can take up the task of training complex deep neural networks. Therefore, researchers have introduced data enhancement techniques to extend the existing datasets. Common data enhancement methods include geometric distortion and photometric distortion. Geometric distortion helps the model learn a wider range of target variations and viewpoints by performing random scaling, rotating, cropping, and panning operations on the image to enhance the model's generalization ability. Photometric distortion, on the other hand, simulates image changes under different lighting conditions and camera settings by changing photometric attributes of the image, such as brightness, contrast, hue, saturation, etc., to enhance the robustness of the model. In addition to the data enhancement methods mentioned above, the POSEIDON data enhancement method was proposed in the literature [43] to address the problem of imbalanced category distribution in the dataset. This method realizes the expansion of the dataset by combining the targets and samples in the original training set to generate new training samples. At the same time, the dataset category imbalance problem is improved. Given that the SDS ODv2 [44] dataset also suffers from category imbalance, we used the POSEIDON method to augment the dataset with data prior to model training. Fig. 1 illustrates the proportion of each category to the total categories in the SDS ODv2 dataset before and after data enhancement, while Fig. 2 provides a comparison of the images before and after data enhancement.



(a) (b)

Fig. 1. Comparison of the proportions of categories in the SDS ODv2 dataset before and after data augmentation. (a) Original dataset. (b) After data augmentation.

III. Our Method

A. Review of FCDS-DETR

Since our proposed solution for object detection applicable to sea rescue missions is constructed based on the results of our previous work, we need to review the basic structure of FCDS-DETR before detailing the specific solution.

As a DETR-like model, FCDS-DETR has a simple structure containing a backbone network, a multilayer Transformer encoder, a multilayer Transformer decoder, a category prediction header, and a bounding box prediction header. Specifically, FCDS-DETR uses ResNet50 [45] as a backbone network to extract feature map $F \in R^{H*W*C}$ from the input image $I \in R^{H0*W0*3}$, where $H0$, $W0$ and H, W denote the height and width of the input image as well as the output feature map, respectively. C denotes the number of feature channels of the output feature image. The feature map F captures feature

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Fig. 2. Example of training set image after data enhancement. The left image is the original image of the SDS ODv2 dataset, and the right image is the data-enhanced image.

information of different levels and complexity in the input image. It is combined with sinusoidal position encoding PE to generate the feature embedding F_{pe} containing spatial position information. The sinusoidal position encoding PE formula is shown in the following equations:

$$\text{PE}\left(\text{pos}, 2i\right) = \text{Sin}\left(\frac{\text{pos}}{10000^{\frac{2i}{d}}}\right), \quad (0 \leqslant i < d/2) \tag{1}$$

$$\text{PE}\left(\text{pos}, 2i+1\right) = \text{Cos}\left(\frac{\text{pos}}{10000^{\frac{2i}{d}}}\right), \quad (0 \leqslant i < d/2) \tag{2}$$

where pos denotes the position coordinates of each pixel point on the feature map in the x- and y-directions, $2i$ and $2i+1$ are the alternating dimensions used to generate sine and cosine components of different frequencies that form the position embedding vector, and d is the total dimension of the position embedding, we use 256 to keep it consistent with the baseline. After that, the F_{pe} will be input into the FCDS-DETR encoder.

The encoder layer of FCDS-DETR inherits the structure of the original DETR, including the multihead self-attention module and the feed-forward network (FFN). The information exchange of feature vectors at all spatial locations is realized by the multihead self-attention module, and the keys (K) and values (V) needed for decoder cross-attention computation are generated.

The core of FCDS-DETR is located in the decoder layers, each of which consists of a multihead self-attention module, a semantic alignment module, a multihead cross-attention module, and an FFN. The initialized query vectors achieve information interaction between potential objects at different spatial locations through multihead self-attention. Subsequently, the output of the multihead self-attention module will be input to the semantic alignment module, whose position in the decoder is shown as the blue module in Fig. 3.

In Fig. 4, we show the internal details of the semantic alignment module. E denotes the feature map obtained by the convolutional transform of the encoder output features. E_{roi} denotes the region of interest containing potential targets extracted by ROI Align [46]. Next, E_{roi} implements feature correction through the CAM module to obtain the output E_{adj} containing the inter-channel dependencies of the feature map. E_{adj} will be used to predict the offset P_{offset_i} of the corresponding sampling point P'_i, where $i = \{0, 1\}$, denoting the group to which the offset belongs, P represents the initial sampling point of the double sampling mechanism, and the offset P_{offset_i} of all sampling points is based on this point. P'_i is the actual sampling position obtained by adding the offset P_{offset_i} to point P. To achieve semantic alignment, the two groups of sampling points will be resampled on the region of interest E_{roi} to generate the content embedding of the new query vector Q'_{cont_i}. The number of sampling points per group on the region of E_{roi} was eight, consistent with the baseline. The coordinates of the sampling points are sinusoidally encoded to generate the spatial position embedding Q'_{pos_i} of the new query vector. The Q'_{cont_i} and Q'_{pos_i} are, respectively, weighted with the output of the multihead self-attention to obtain two sets of query vectors Q'_i. These two sets of query vectors will be used as inputs to perform the multihead cross-attention computation in parallel. The multihead cross-attention (MHCAttention) representation is shown in the following equation:

$$\begin{aligned}&\text{MHCAttention}\left(Q'_i, K, V\right)\\ &= \text{Concat}\left(\text{Softmax}\left(\frac{Q'_{i,j}K_j}{\sqrt{d_k}}\right)V_j\right)W^o\end{aligned} \tag{3}$$

where $Q'_{i,j}$, K_j, and V_j denote the jth feature groups of Q'_i, K, and V, usually j takes 8. d_k is the row dimension of $Q'_{i,j}$ and K_j. W^o represents the output transformation matrix. The results of attention computation for multiple feature groups are stitched together to obtain the two attention weight maps F_{wi} corresponding to Q'_i. Subsequently, the two attention weight maps are superimposed to obtain the final attention result. Finally, the object detection process is realized by the category prediction header and the bounding box prediction header.

FCDS-DETR's core principle is to apply channel attention and double sampling to the encoder output features, producing two sets of new query vectors. These query vectors are situated in the same embedding space as the encoder output feature map, ensuring semantic alignment. This alignment enhances the accuracy of potential target region extraction and, as a result, improves the model's detection accuracy.

B. Overall Structure

Since many small targets and complex backgrounds characterize remote sensing images at sea, it is not appropriate to directly apply FCDS-DETR, an object detector for generic scenarios, to this context. To cope with this problem, we make two key improvements to the decoder part of FCDS-DETR. First, we add a 2-D spatial Gaussian probability density distribution into the cross-attention module to regulate the search range of the query vectors in cross-attention to improve the model's ability to quickly localize targets in complex marine scenes. Second, we incorporate the general denoising training strategy from DN-DETR [47] into FCDS-DETR.

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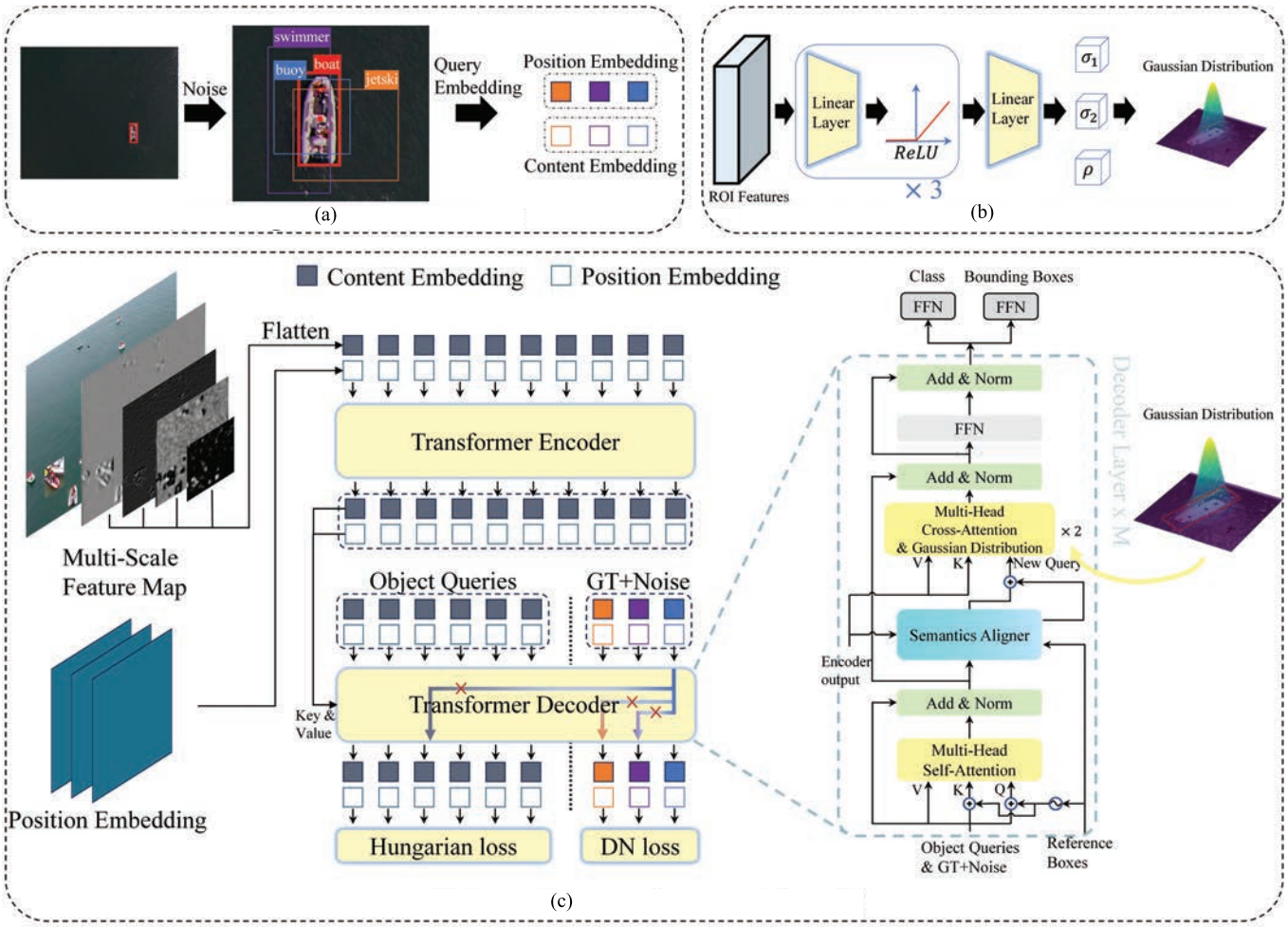


Fig. 3. (a) Process of adding noise to ground-truth bounding boxes and labels. (b) Generation mechanism of the discrete coefficients (σ_1, σ_2) along the height and width of the feature map, as well as the correlation coefficient (ρ) for the center of the Gaussian distribution. (c) Overall model architecture, including the encoder, decoder, classification head, and bounding box prediction head. A 2-D Gaussian probability density distribution is used in the decoder to regulate the query search region in cross-attention, enabling efficient localization of potential targets. A denoising module is introduced to accelerate convergence and enhance robustness.

By adding noise to the ground-truth bounding boxes and performing a denoising task to reconstruct the corresponding ground-truth boxes, this approach reduces the difficulty of bipartite matching and helps the model converge more quickly. This method helps to promote the convergence of the model without increasing the number of parameters in the inference stage of the model. The overall structure of the model is shown in Fig. 3. Next, this article will discuss in detail the application of Gaussian probability density distribution in 2-D space, the fusion of Gaussian attention and multihead cross-attention, and the denoising training method.

C. Applications of 2-D Gaussian Probability Density Distributions

Remote sensing images usually have a high resolution to show more scene details. However, in remote sensing images, the trapped people to be detected usually occupy only a dozen or tens of pixels, and the area of their target pixels accounts for less than one-thousandth of the whole image area, which belongs to tiny targets. It is a great challenge to make the model quickly focus on the suspicious regions where these tiny targets exist during the training and inference process. When using the FCDS-DETR model for detection in generic scenarios, in the initial stage, multiple decoder layers can only rely on multihead cross-attention to search for tiny targets in the image, resulting in the model needing to spend a lot of time focusing on the localization of the suspicious positions of the targets in the early stage of training. This is unfavorable for tiny target object detection in maritime scenarios. Inspired by the "center-surround" perception mechanism of human vision [48], and drawing on the advantages of the 2-D Gaussian distribution in terms of rotational symmetry, exponential decay, and parameter separability, we introduce a 2-D spatial Gaussian probability density distribution in each decoder layer. This form of distribution fits the need for modeling target localization uncertainty in computer vision. Through this mechanism, we were able to dynamically adjust the spatial search range of the query vectors in the cross-attention, thus guiding the model to focus on suspicious regions in the image where tiny targets may be present at an early stage, effectively improving the detection accuracy.

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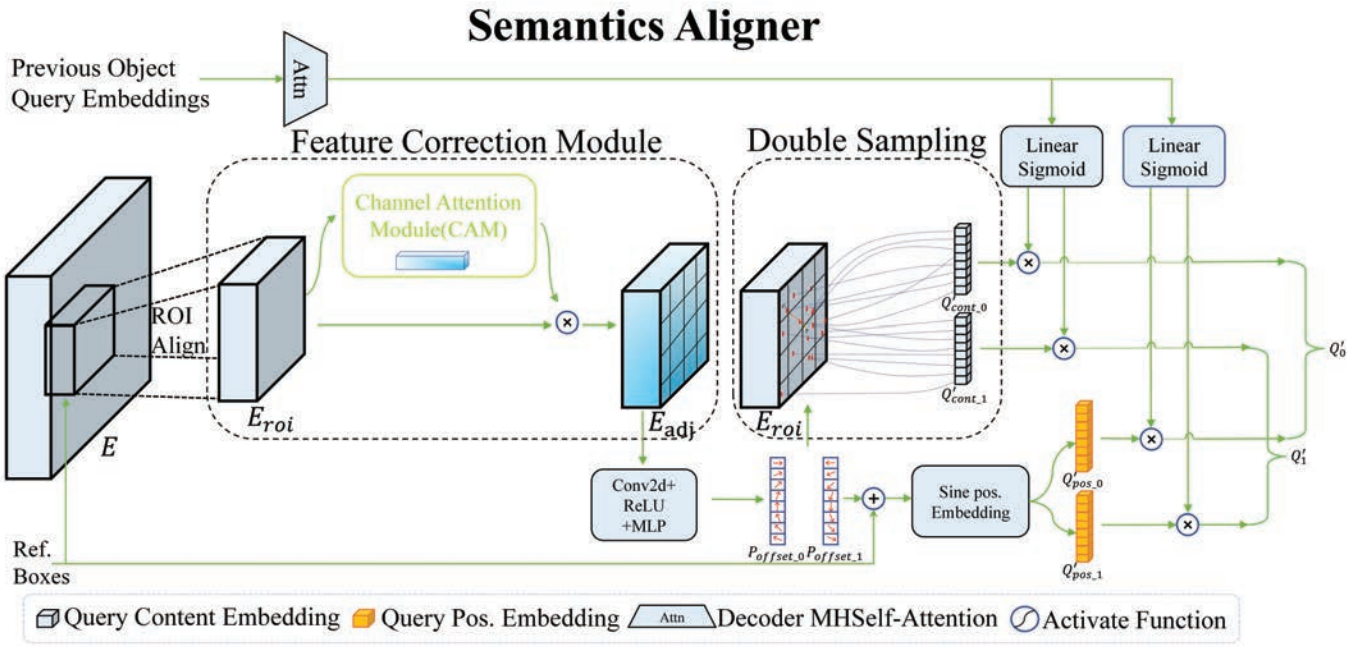


Fig. 4. Overall structure of the semantic aligner, including the feature correction module and the double sampling mechanism. The feature correction module improves the sampling point localization accuracy by explicitly modeling the dependencies between the feature channels to be sampled. The double sampling mechanism samples over the feature region E_{roi} to generate inputs for multiple cross-attention. The yellow point in the E_{roi} feature indicates the center of the ROI region, while the red points indicate the actual sampling locations.

Specifically, we take the resampling point coordinates obtained in the semantic alignment module as the mean μ_1 and μ_2 of the Gaussian probability density distribution in 2-D space, generating multiple Gaussian probability distributions centered on the resampling points. These Gaussian probability distributions will act as additional attentional mechanisms for restricting attention to the neighborhood of the initial location of the object query. This allows the model to quickly perceive the target's position in the image instead of searching blindly in all regions. It ensures that the features that the model focuses on are more relevant to the final predicted target object. Also, we designed an MLP network as shown in Fig. 3(b) that uses the features extracted by ROI Align as inputs for predicting the dispersion coefficients σ_1 and σ_2 of each Gaussian probability distribution over the height and width of the feature map and the correlation coefficients ρ of the distribution locations. The 2-D Gaussian probability density distribution representation is shown in the following equation:

$$G(x,y) = \exp\left[-\frac{1}{\alpha}\left(\frac{(x-\mu_1)^2}{\sigma_1^2} - \frac{\rho(x-\mu_1)(y-\mu_2)}{\sigma_1\sigma_2} + \frac{(y-\mu_2)^2}{\sigma_2^2}\right)\right] \quad (4)$$

where α is the hyperparameter that regulates the bandwidth of the Gaussian distribution, and x and y are the center coordinates of each pixel of the feature map. $(x,y) \in [(1/2),(w-1/2)] \times [(1/2),(h-1/2)]$. In the generated Gaussian weight map G, the regions closer to μ_1 and μ_2 will obtain higher importance, and vice versa, the importance will be lower. Fig. 5 shows the 3-D image of the generated 2-D Gaussian probability distribution at the sampling points.

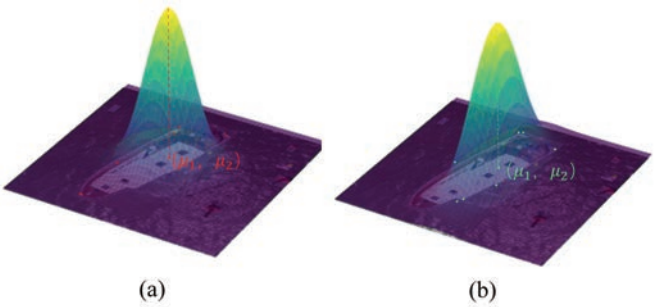


Fig. 5. 3-D image of a 2-D Gaussian probability distribution. (a) Gaussian distribution corresponding to the resampling point when $i = 0$. (b) Gaussian distribution corresponding to the resampling point when $i = 1$.

D. Fusion of Gaussian Attention and Multihead Cross-Attention

After obtaining the Gaussian probability density distribution weight map G in 2-D space, we need to fuse G with the attention weight map F_{wi} to realize the spatial modulation of the attention weight map. According to Section III-A, it can be seen that inputting two sets of sampling points P'_0 and P'_1 in parallel into the multihead cross-attention module for computation will result in attention weight maps F_{w0} and F_{w1}. F_{w0} denotes the result of multihead cross-attention calculation between the new query vector Q'_0 corresponding to the 0th group sampling points and the encoder output features; F_{w1} denotes the result of multihead cross-attention calculation between the new query vector Q'_1 corresponding to the 1st group sampling points and the encoder output features. When performing Gaussian spatial modulation on the weight map,

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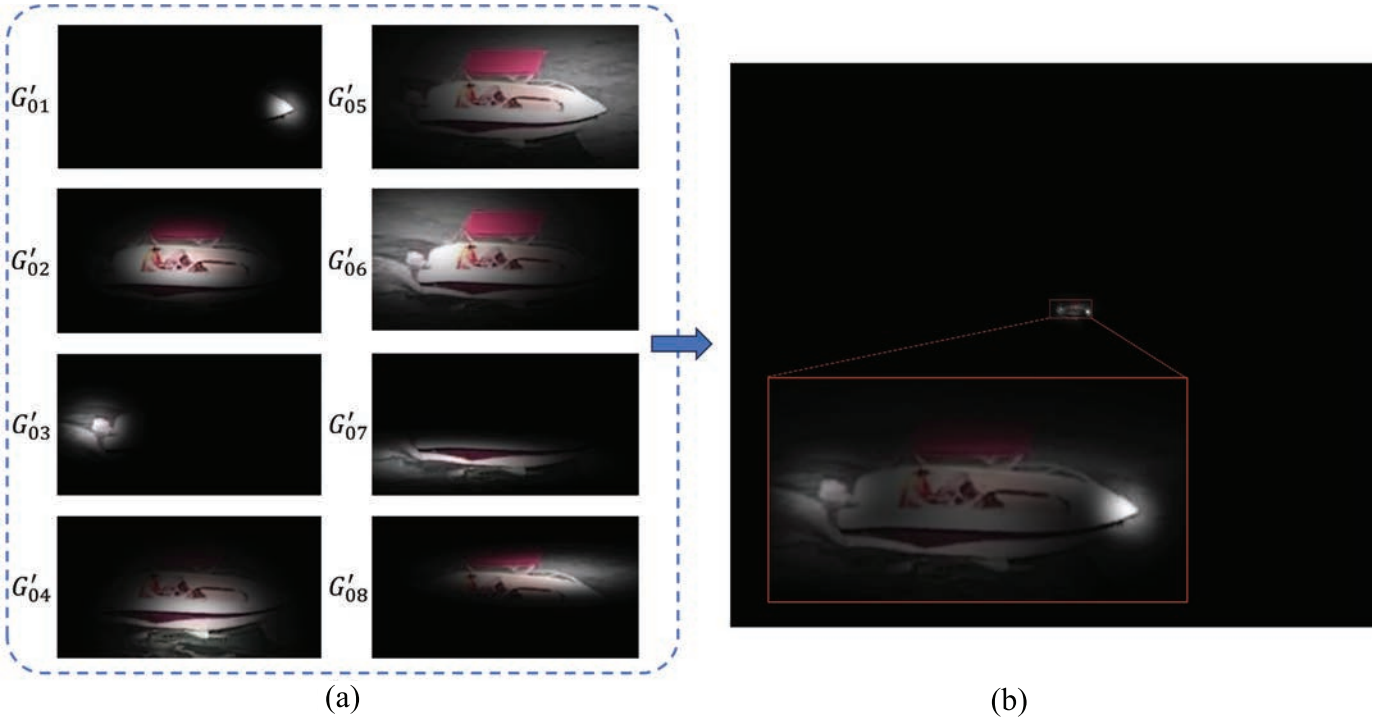


Fig. 6. Effect of 2-D Gaussian probability density distribution on image. (a) 2-D Gaussian probability density distribution at each resampling point. (b) Combined effect of the 2-D Gaussian probability distributions at the eight sampling points.

we make the Gaussian probability density distribution shared within groups and not shared between groups. In FCDS-DETR, the number of sampling points in each group is 8, so the set of Gaussian probability distributions we obtain for the two groups can be expressed as follows:

$$G_i{}' = \{G_{i1}, G_{i2}, G_{i3}, \ldots, G_{i8}\} \quad i = \{0, 1\} \tag{5}$$

where G'_0 and G'_1 denote the set of Gaussian probability density distributions corresponding to each of the two sets of sampling points.

Subsequently, each independent Gaussian probability distribution set G'_i generated by the ROI Align region is added to the corresponding cross-attention of the group, and then normalized by Softmax to obtain the spatial modulation result C_i. G'_i modulates the formula for a multihead cross-attention map, as shown in the following equation:

$$C_i = \text{Concat}\left(\text{Softmax}\left(\frac{Q'_{i,j}K_j}{\sqrt{d_k}} + G'_i\right)V_j\right) \tag{6}$$

where C_i denotes the result of Gaussian spatial modulation of the weight map by each set of Gaussian probability density distributions G'_i, from (6), multiple attention heads j in the ith group of cross-attention share the 2-D spatial Gaussian probability density distribution G'_i of the group. With this modulation, the result C_i of the spatial modulation is made to exhibit greater cross-attention weights close to the ROI Align region. Fig. 6 illustrates the impact of a 2-D Gaussian probability density distribution on an image when $i = 0$, where subplot (a) depicts the distribution at each resampling point and subplot (b) shows the combined effect of these distributions at eight sampling points when $i = 0$.

E. Denoising Training

Both FCDS-DETR and the traditional object detector based on Transformer architecture adopt the bipartite graph matching algorithm in the decoder part, which is used to achieve a one-to-one matching between the ground-truth bounding box and the predicted bounding box. However, during the training process of the model, the bipartite graph matching algorithm suffers from the disadvantage of unstable matching, which leads to the model needing a large amount of time to stabilize the bipartite graph matching process in the early training stage, resulting in slow convergence of the model. To solve this problem, we combine the denoising training method [47] with the FCDS-DETR decoder to improve the convergence speed of the model.

Specifically, in each image of the training dataset, there are one or more real targets to be detected. Each ground-truth target is labeled by a bounding box and a category label. The denoising training method stabilizes the bipartite graph matching by introducing random noise into the bounding boxes and category labels, such as GT + Noise shown in Fig. 3(c). The method forces the model to reconstruct the ground-truth bounding box in the bounding box to which the noise has been added as a way to stabilize the bipartite graph matching.

The addition of noise on the bounding box consists of both center shifting of the bounding box and random scaling of the bounding box. The true bounding box for each target is represented by a tuple (x, y, w, h) uniquely determined, where x and y denote the center coordinates of the bounding box; w and h denote the corresponding width and height of each bounding box. The denoising training method introduces a

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variable $\beta \in (0, 1)$ to control the scale of the added noise. For the center shift of the bounding box, we get the new center coordinates (x', y') by adding random noise $(\Delta x, \Delta y)$ to the center coordinates and making it satisfy the following equation:

$$\begin{cases} x' = x + \Delta x, & |\Delta x| < \frac{\beta w}{2} \\ y' = y + \Delta y, & |\Delta y| < \frac{\beta h}{2}. \end{cases} \tag{7}$$

This ensures that the center coordinates of the bounding box remain within the original bounding box after adding the noise. For random scaling of the bounding box, the variable β is used to control the scaling range, such that the width w' and height h' of the scaled bounding box satisfy the following equation:

$$\begin{cases} w' = w + \Delta w, & |\Delta w| \le \beta w \\ h' = h + \Delta h, & |\Delta h| \le \beta h. \end{cases} \tag{8}$$

This ensures that the scaled bounding box is 0–2 times the original bounding box. For noise addition on category labels, a variable $\gamma \in (0, 1)$ is introduced to control the proportion of category labels that are randomly flipped, ensuring that the ground-truth label is flipped to a label other than itself. The purpose of label flipping is to force the model to predict the ground-truth class based on the noise box (x', y', w', h') in order to capture the potential relationship between the ground-truth box and the label in a more meaningful way.

Through the appeal method, we can obtain multiple sets of noisy bounding boxes and category labels, as shown in Fig. 3(a). Subsequently, we use the noisy bounding box as the position embedding of the query vector in the decoder and the category label as the content embedding of the query vector in the decoder to denoise the model training. In addition, we inherit the attention mask mechanism in DN-DETR [47] to prevent information leakage from the denoising part to the matching part and avoid information leakage between different noise types of the same real target.

Since the denoising module is applied only in the training phase of the model, no additional parameters are introduced during the inference process of the model, and there is no impact on the model's inference speed.

IV. Experiments

A. Dataset

This study conducted experiments on the SDS ODv2 [44] dataset and the AFO [49] dataset. The purpose of the SDS ODv2 dataset is to assist in developing UAV algorithms for maritime object detection. The dataset contains 32 495 images (27 198 training images, 1547 validation images, and 3750 test set images) enhanced by the POSEIDON method. All images were captured by various types of UAVs operating at low altitudes (ranging from 5 to 260 m) with viewing angles adjusted between 0° and 90°. Five object types are to be detected in the dataset, including swimmer, boat, jetski, life_saving_appliances, and buoy. The AFO dataset is the first freely available dataset to be used for training maritime SAR missions and contains more than 40 000 objects to be detected. All images in the dataset were taken by UAVs at low altitudes from 30 to 80 m. The dataset consists of six object detection categories: human, wind\ sup-board, boat, buoy, sailboat, and kayak, each labeled using a bounding box and a category label.

TABLE I
Evaluation Indicators (↑ Means Bigger is Better, and ↓ Means Smaller is Better)

Index	Description
AP↑	Average precision at IoU = 0.50 : 0.05 : 0.95
AP_{50}↑	Average precision at IoU=0.50
AP_{75}↑	Average precision at IoU=0.75
AR_1↑	Average Recall rate given 1 detections per image
AR_{10}↑	Average Recall rate given 10 detections per image

B. Experimental Setup

Our training and inference device includes an Xeon Silver 4210 2.2 GHz CPU, 10TB of ROM, and 4× NVIDIA GeForce RTX 3090 GPUs. The model undergoes training for 100 epochs using both the SDS ODv2 dataset and the AFO dataset. During the training of the model, ResNet-50 was used as the feature extraction network, and the initial learning rate of the model was set at 1×10^{-5}. The learning rate decreases to 1/10 of the initial value after 80 epochs and is optimized using the AdamW optimizer. The batch size is set to 1. In the experiment, the hyperparameter α was set to 5, which ensured that the 2-D Gaussian probability density distribution had a large receptive field at the beginning of training, covering a wider spatial range and thus obtaining richer information gradients.

C. Evaluation Indicators

We use the COCO evaluation metric to evaluate the detection performance of each model objectively. Among them, average precision (AP) is the most widely used evaluation metric in the field of object detection. It is obtained by averaging multiple precision values on a *PR* curve with precision as the vertical axis and recall as the horizontal axis. Precision rate is the proportion of samples predicted as positive by the model that were correctly predicted as positive, while recall rate is the proportion of samples actually predicted as positive by the model. Before calculating the AP, the samples need to be divided according to the combination of the true class and the predicted class of the learner so as to form true positive (TP), false positive (FP), true negative (TN), and false negative (FN). The precision rate P and recall rate R are calculated as follows:

$$P = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{9}$$

$$R = \frac{\text{TP}}{\text{TP} + \text{FN}}. \tag{10}$$

The COCO metrics and their meanings are presented in Table I.

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TABLE II
COMPARISON OF THE PERFORMANCE ON THE SDS ODV2 DATASET

Model	Backbone	Epochs	Params(M)	FPS	GFLOPs	AP	AP_{50}	AP_{75}	AR_1	AR_{10}	Improvement(AP)
Faster-RCNN [17]	VGG16	100	34	15	118	24.5	52.1	20.4	23.6	32.1	+26.5%
YOLOv5 [50]	CSPDarknet-53	100	7	33	16	33.7	56.9	35.1	29.5	39.7	+17.3%
DETR [25]	ResNet-50	100	41	9	104	14.0	28.2	12.3	17.2	28.8	+37.0%
Deformable-DETR [36]	ResNet-50	100	32	12	90	35.7	69.4	33.8	28.5	45.6	+15.3%
Conditional-DETR [51]	ResNet-50	100	43	8	94	25.5	56.5	19.0	22.4	37.1	+25.1%
YOLOv7 [20]	ResNet-50	100	37	34	105	44.3	76.8	45.0	38.5	51.6	+6.7%
YOLOv12 [52]	R-ELAN	100	59	6	185	45.0	77.2	-	-	62.5	+6.0%
RT-DETR [53]	ResNet-50	100	43	41	136	44.8	76.5	-	-	57.2	+6.2%
Baseline	ResNet-50	100	61	5	190	45.3	71.2	48.5	37.5	52.1	+5.7%
Ours	ResNet-50	**100**	**62**	5	**190**	**51.0**	**82.7**	**54.4**	**37.0**	**59.9**	-

TABLE III
COMPARISON OF THE PERFORMANCE ON THE AFO DATASET

Model	Backbone	Epochs	Params(M)	FPS	GFLOPs	AP	AP_{50}	AP_{75}	AR_1	AR_{10}	Improvement(AP)
Faster-RCNN [17]	VGG16	100	34	15	118	27.5	54.4	25.0	23.8	30.3	+27.0%
YOLOv5 [50]	CSPDarknet-53	100	7	34	16	37.8	73.5	38.5	13.3	30.9	+16.7%
DETR [25]	ResNet-50	100	41	9	104	11.4	30.9	7.0	7.4	22.3	+43.1%
Deformable-DETR [36]	ResNet-50	100	32	11	90	34.9	64	35.7	20.9	42.7	+19.6%
Conditional-DETR [51]	ResNet-50	100	43	9	94	20.7	47.2	14.0	15.4	34.3	+33.8%
YOLOv7 [20]	ResNet-50	100	37	35	105	42.0	76.7	40.3	33.2	52.3	+12.5%
YOLOv12 [52]	R-ELAN	100	59	5	185	44.2	76.5	-	-	66.8	+10.3%
RT-DETR [53]	ResNet-50	100	43	39	136	40.1	74.1	-	-	65.3	+14.4%
Baseline	ResNet-50	100	61	6	190	50.1	81.4	53.3	36.7	58.7	+4.4%
Ours	ResNet-50	**100**	**62**	5	**190**	**54.5**	**87.5**	**61.6**	**26.9**	**57.5**	-

D. Experimental Results

In this section, we assess the model's detection performance on both the SDS ODv2 dataset and the AFO dataset, comparing it with other conventional object detection methods. Tables II and III present the experimental results for the model on the respective datasets. As observed, the baseline model, with its strong performance, outperforms other Transformer-based comparison models on the SDS ODv2 and AFO datasets, achieving average accuracies of 45.3% and 50.1%, respectively. However, building upon the baseline model, our improved model further enhances the accuracy, with average accuracies of 51.0% and 54.5% on the two datasets, significantly surpassing both the baseline model and other detection models.

Based on the network structure of the original FCDS-DETR, the proposed object detector suitable for maritime rescue missions only adds a 2-D Gaussian prediction network and a denoising training module, so the increase in the number of model parameters is limited. As shown in Table II, compared to the baseline model, the number of model parameters on the SDS ODv2 dataset is only increased by 1.61% but gains a significant advantage in the detection performance (+5.7% AP). Compared with the classic two-stage object detector Faster R-CNN, the model's AP improved by 26.5%; compared with the real-time end-to-end detector RT-DETR, the AP improved by 6.2%. In addition, compared to the widely used one-stage object detectors YOLOv5, YOLOv7, and YOLOv12, the AP of the model is improved by 17.3%, 6.7%, and 6.0% respectively, which shows that the Transformer-based object detector is able to more accurately detect targets by more comprehensively taking into account and understanding the global contextual information of the image compared to the two-stage or one-stage object detectors. Fig. 7 presents a comparison of detection results on the SDS ODv2 dataset between our proposed model, the single-stage detector YOLOv12, and the baseline model. Specifically, the first column shows the results of the YOLOv12 model, the second column displays the results of our proposed model, and the third column presents the results of the baseline model. It can be observed that our proposed model is capable of detecting more maritime objects under varying flight altitudes, camera tilt angles, and sea surface conditions, compared to both the baseline model and YOLOv12. While YOLOv12 is more sensitive to large maritime targets such as boats and jet skis, it shows lower sensitivity to swimmers due to their minimal pixel occupancy. As shown in the dashed box areas indicated by the orange

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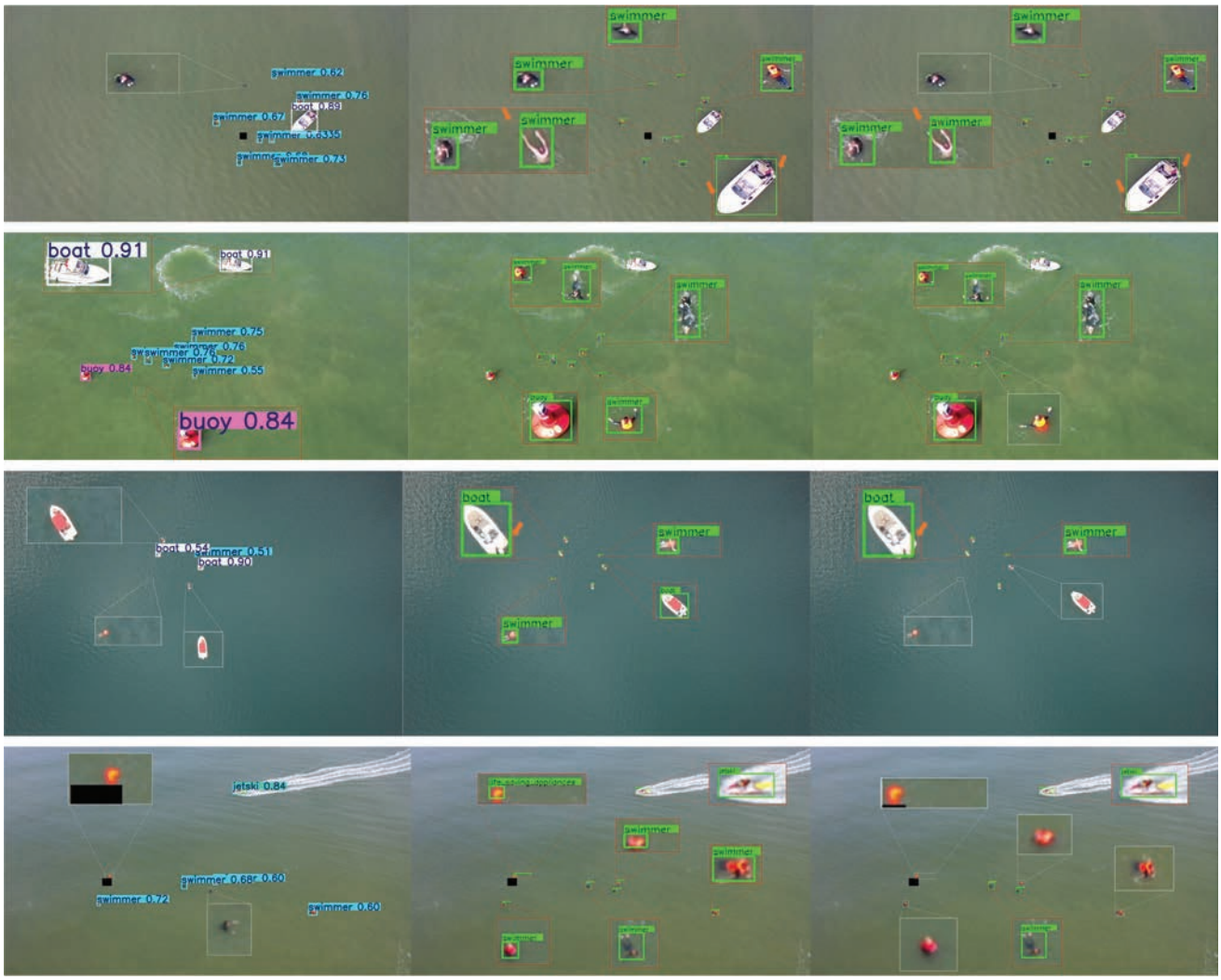


Fig. 7. Comparison of detection results on the SDS ODv2 dataset. The first column presents the results obtained by the YOLOv12 model; the second column shows the results of our proposed model in maritime rescue scenarios; and the third column displays the results of the baseline model. The orange arrow points to a detailed comparison of the two model detection results.

arrows in the figure, compared to the baseline model, the incorporation of a 2-D Gaussian probability density distribution as an auxiliary attention mechanism significantly enhances the model's sensitivity to object boundaries and improves localization accuracy. This fully demonstrates the performance advantages of our system in complex scenarios and aligns with the expected outcomes of our model design. We attribute the improvement in the model's localization performance to the fact that the addition of Gaussian attention limits the search range of the query vectors in cross-attention, which enables the model to quickly localize the target to be detected in the image at an early stage.

On the AFO dataset, our model continues to exhibit outstanding performance. According to the data in Table III, the detection accuracy of the model improves by 4.4% relative to the baseline model. Compared to Faster-RCNN and RT-DETR, the AP improved by 27.0% and 14.4%, respectively. Compared to YOLOv5, YOLOv7, and YOLOv12, our model also achieved notable improvements, increasing AP by 16.7%, 12.5%, and 10.3%, respectively. In terms of detection speed, since the average resolution of images in the SDS ODv2 and AFO datasets exceeds 2K, the maritime object detector proposed in this article has a slightly higher computational load than the YOLO series and RT-DETR real-time detection models that use a fixed input size (default 640 $\times$ 640), resulting in slightly lower FPS performance than the above methods. Fig. 8 shows the detection results on the AFO dataset. It can be observed that our model continues to exhibit strong robustness on this dataset. In particular, under complex background conditions such as white splashes caused by swimmers' movements, our model achieves more accurate target localization compared to the YOLOv12 model. Additionally, the regions indicated by the orange arrows illustrate a comparison of localization accuracy between our model and FCDS-DETR in single-target detection scenarios. Compared with the generalized object detector FCDS-DETR, the maritime rescue object detection model proposed in this article can not only detect more maritime targets but also localize the targets more accurately. Meanwhile, for detecting multiple overlapping objects in the AFO dataset, the model proposed in this article also performs better than the baseline model. In addition, the detection results on two different datasets show that our model exhibits

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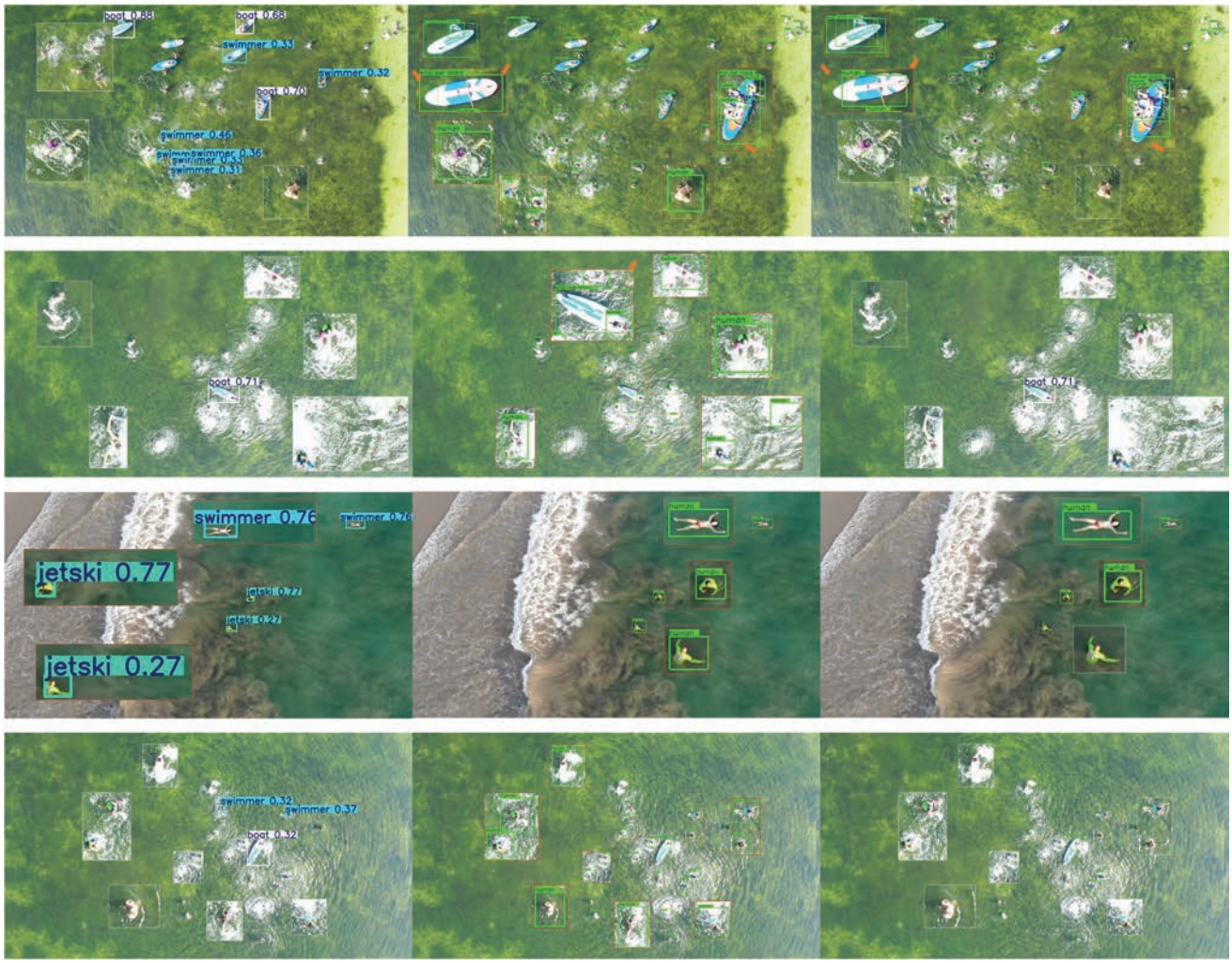


Fig. 8. Comparison of detection results on the AFO dataset. The first column presents the results obtained by the YOLOv12 model; the second column shows the results of our proposed model; and the third column displays the results of the baseline model. The orange arrow points to a detailed comparison of the two model detection results.

TABLE IV
ABLATION EXPERIMENTS ON THE SDS ODV2 DATASET

Row	Params(M)	CAM	Double Sampling	2D Gauss	GT+Noise	AP	AP_{50}	AP_{75}	Improvement(AP)
1	61	√	√			45.3	71.2	48.5	-
2	62	√	√	√		48.0	77.6	51.1	+2.7
3	62	√	√	√	√	51.0	82.7	54.4	+5.7
4	58	√		√		46.2	78.5	46.8	+0.9
5	55	√			√	46.5	79.1	47.8	+1.2
6	61		√	√		47.6	79.0	49.6	+3.4
7	60		√		√	48.2	81.4	49.0	+2.9

good adaptability and generalization ability in similar maritime object detection tasks, with the potential to be generalized to other similar application scenarios.

E. Ablation Analysis

In order to validate the impact of 2-D Gaussian probability density distribution and denoising training methods on the model detection performance, we designed ablation experiments on the SDS ODv2 dataset and compared the results with the baseline FCDS-DETR. As shown in the first three rows of Table IV, the first row corresponds to the baseline model FCDS-DETR, while the second and third rows represent the models obtained by successively introducing the 2-D Gaussian probability density distribution and the denoising training

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strategy into the baseline. From the first to the third row, AP/AP50/AP75 gradually increases from 45.3 /71.2 /48.5 to 51.0 /82.7 /54.4.

1) Effectiveness of the 2-D Gaussian Probability Distribution: The Gaussian probability density distribution is added as an additional attention mechanism based on the original cross-attention map. It aims to limit the attention near the initial position of the object query to improve the model's ability to perceive the target position in the image. At the same time, the attention fusion method described in Section III-D is used to adapt to the attention map fusion problem caused by multiple groups of sampling points. It is worth noting that traditional pooling-based spatial attention mechanisms typically apply operations such as max pooling to compress the channel dimensions of the ROI, thereby generating spatial attention maps. However, such methods retain only the local maxima in the features, leading to the loss of fine spatial details. This issue is particularly pronounced when detecting small targets in complex backgrounds, as these methods may overlook the regions where such targets reside. In contrast, the 2-D Gaussian probability density approach proposed in this article introduces a continuous and scale-adaptive Gaussian prior to each sampling point corresponding to a candidate target. This prior softly modulates the cross-attention weights, enabling the attention mechanism to adaptively focus around the predicted initial locations. As a result, the method significantly enhances the preservation of spatial information and improves the detection sensitivity to small targets. The results are shown in Table IV. Compared to the baseline model, the improved model with the inclusion of a 2-D Gaussian probability distribution improves the detection precision by +2.7%. The results show that the introduction of a 2-D Gaussian probability distribution is effective in improving the detection accuracy of the model.

2) Effectiveness of Denoising Training: As shown in Table IV, we use the denoising training method for the model that incorporates a 2-D Gaussian probability distribution to force the model to reconstruct the ground-truth bounding box from the noise-containing bounding box and predict the class of the target. A comparison with the baseline model shows that the improved model using the denoising training method improves the detection AP by +3.0% for the same number of epochs, which is a very impressive result. This shows that the use of the denoising training method is effective in improving object detection accuracy in sea rescue scenarios.

3) Performance Impact Analysis of Module Combination Strategies: In addition, to further investigate the impact of different module combinations on model performance, we conducted ablation experiments by combining the CAM and double sampling mechanism introduced in FCDS-DETR with the 2-D Gaussian probability density distribution and denoising training method proposed in this article, respectively. The results are shown in rows 4–7 of Table IV. The experimental results of CAM and 2-D Gaussian indicate that, even without denoising training, the model can achieve a 0.9% improvement in AP by relying solely on these two components, indicating that even without a denoising mechanism, this combination still yields performance gains. The results in row 5 indicate that the denoising training method has independent effectiveness, significantly improving model performance even without relying on 2-D Gaussian modeling. The results in rows 6 and 7 show that the double sampling mechanism, when combined with either 2-D Gaussian modeling or the denoising training method, significantly improves model performance, with AP increases of +3.4% and +2.9%, respectively, outperforming other combinations without these components.

V. Conclusion

Remote sensing images captured by UAVs in sea rescue scenarios are characterized by small rescue targets and high background interference, which brings great challenges to general object detectors. To address this problem, this article proposes a novel object detector based on the Transformer architecture and obtains excellent performance. The core idea of this study is to introduce a 2-D Gaussian probability density distribution and a denoising training method, which enables the model to be better adapted to the task of object detection in maritime scenarios. Eventually, the effectiveness of the model is proved through a large number of experiments. We hope that this research can bring help to object detection tasks, object tracking tasks, and semantic segmentation tasks based on maritime UAV platforms, specifically including maritime search and rescue, marine environment monitoring, maritime traffic management, and UAV mapping. In addition, due to the generality of the Gaussian attention method proposed in this article, subsequent researchers can apply it to the feature map processing of their models so as to improve the model performance and explore its potential in different tasks. It is worth noting that all currently available public maritime target detection datasets have been collected at low altitudes (below 500 m). Therefore, the performance of our model on datasets captured at altitudes above 500 m remains to be explored.

As a current limitation of our model, it is not yet optimized for deployment on embedded edge computing devices. Since maritime rescue missions are highly time-sensitive, future work will focus on tailoring the existing model to make it suitable for resource-constrained platforms carried by UAVs, while ensuring detection performance to further enhance detection efficiency.

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